

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276583

Kind Code

A1

Publication Date

September 04, 2025

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ELECTRIC VEHICLE

Abstract

The electric vehicle according to the present disclosure calculates motor torque using an MT vehicle model simulating an MT vehicle having a manual transmission and an internal combustion engine. In the first operation mode, an operation amount of a pseudo-clutch pedal and a shift position of a pseudo-gearshift are input to the MT vehicle model to reflect operation of the pseudo-clutch pedal and operation of the pseudo-gearshift in electric motor control. In the second operation mode where the operation of the pseudo-clutch pedal is not needed, an operation amount of a clutch pedal calculated by a driver model is input to the MT vehicle model instead of the operation amount of the pseudo-clutch pedal.

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Appl. No.: 19/210643

Filed: May 16, 2025

Foreign Application Priority Data

JP	2020-135122	Aug. 07, 2020
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Related U.S. Application Data

parent US continuation 18799020 20240809 parent-grant-document US 12337700 child US 19210643

parent US continuation 18437905 20240209 parent-grant-document US 12090861 child US

Publication Classification

Int. Cl.: **B60L15/20** (20060101); **B60K1/00** (20060101); **B60K26/02** (20060101); **G06F30/20** (20200101); **G06F111/10** (20200101); **H02P6/08** (20160101)

U.S. Cl.:

CPC **B60L15/20** (20130101); **B60K26/02** (20130101); **G06F30/20** (20200101); **H02P6/08** (20130101); B60K2001/001 (20130101); B60K2026/025 (20130101); B60L2250/00 (20130101); B60L2260/24 (20130101); B60L2260/46 (20130101); G06F2111/10 (20200101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This is a Continuation of U.S. patent application Ser. No. 18/799,020 filed Aug. 9, 2024, which in turn is a Continuation of U.S. patent application Ser. No. 18/437,905 filed Feb. 9, 2024 (now U.S. Pat. No. 12,090,861), which is a Continuation of U.S. patent application Ser. No. 17/349,406 filed Jun. 16, 2021 (now U.S. Pat. No. 11,932,118), which claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2020-135122, filed Aug. 7, 2020. The contents of the prior applications identified above are incorporated herein by reference in their entireties.

BACKGROUND

Field

[0002] The present disclosure relates to an electric vehicle configured to use an electric motor as a power device for traveling.

Background Art

[0003] An electric motor used as a power device for traveling in an electric vehicle differs greatly in torque characteristic from an internal combustion engine used as a power device for traveling in a conventional vehicle. Due to the difference in torque characteristics of power devices, a transmission is essential for the conventional vehicle, whereas in general the electric vehicle is not equipped with a transmission. Of course, the electric vehicle is not equipped with a manual transmission (MT: Manual Transmission) that switches a gear ratio by manual operation by a driver. Therefore, there is a great difference in a driving feeling between driving of the conventional vehicle with the MT (hereinafter referred to as MT vehicle) and driving of the electric vehicle.

[0004] On the other hand, the torque of the electric motor can be controlled relatively easily by controlling the applied voltage and magnetic field. Therefore, the electric motor can obtain a desired torque characteristic within an operating range of the electric motor by implementing appropriate motor control. Taking advantage of this feature, a technique to simulate the torque characteristic peculiar to the MT vehicle by controlling the torque of the electric vehicle has been proposed so far.

[0005] JP 2018-166386 discloses a technique for producing a pseudo shift change in a vehicle that transmits torque to wheels from a drive motor. In this vehicle, at a predetermined opportunity defined by a vehicle speed, an accelerator opening, an accelerator opening speed, or a brake

depression amount, after reducing the torque of the drive motor by a set variation amount, torque variation control is performed to increase the torque again at a predetermined time period. Thus, an uncomfortable feeling given to a driver familiar with a vehicle equipped with a stepped transmission is suppressed.

[0006] However, in the above technique, it is impossible to determine the timing of executing the torque variation control simulating the speed change operation voluntarily by the driver's own operation. In particular, for the driver accustomed to driving the MT vehicle, pseudo speed change operation without intervention of manual speed change operation by the driver has a possibility that a discomfort is given to the driving feeling of the driver seeking pleasure to operate the MT.

SUMMARY

[0007] In view of such circumstances, the inventors of the present application are considering providing a pseudo-gearshift and a pseudo-clutch pedal on the electric vehicle so as to obtain a feeling of driving the MT vehicle in the electric vehicle. Of course, these pseudo-devices are not simply attached to the electric vehicle. The inventors of the present application are considering allowing the electric motor to be controlled by operating the pseudo-gearshift and pseudo-clutch pedal so that the torque characteristic similar to that of the MT vehicle can be obtained.

[0008] However, it is also a fact that driving the MT vehicle involves peculiar difficulties. In particular, the operation of the clutch pedal is sometimes cumbersome and difficult operation for a modern driver familiar with driving an AT vehicle with an automatic transmission (AT). For example, it can be said that clutch operation in reverse, clutch operation in hill starting, clutch operation in a sharp curve, clutch operation in a traffic jam, clutch operation in a railroad crossing, clutch operation in a garage, and clutch operation on a gravel road are particularly difficult operation. This is a similarly possible problem when simulating the driving feeling of the MT vehicle in the EV. If the operation required by the MT vehicle is applied to the EV as it is, even the difficulty peculiar to the MT vehicle will be simulated, and the driver may not be able to enjoy the driving feeling like the MT vehicle.

[0009] The present disclosure has been made in view of the above problems, and an object thereof is to provide an electric vehicle capable of enjoying a driving feeling like an MT vehicle without experiencing difficulties peculiar to the MT vehicle.

[0010] The electric vehicle according to the present disclosure is an electric vehicle using an electric motor as a power device for traveling, comprising an accelerator pedal, a pseudo-clutch pedal, a pseudo-gearshift, an operation mode selector, and a controller. The operation mode selector is a device configured to switch an operation mode between a first operation mode and a second operation mode. The first operation mode is an operation mode reflecting operation of the pseudo-clutch pedal and operation of the pseudo-gearshift in control of the electric motor. The second operation mode is an operation mode not requiring the operation of the pseudo-clutch pedal for the control of the electric motor. The controller is a device configured to control a motor torque output by the electric motor in accordance with the control mode selected by the mode selector.

[0011] The controller comprises a memory and a processor. The memory stores an MT vehicle model and a first driver model. The MT vehicle model is a model simulating a torque characteristic of a driving wheel torque in an MT vehicle. The MT vehicle referred to herein is a vehicle having an internal combustion engine whose torque is controlled by operation of a gas pedal and a manual transmission whose gear stage is switched by operation of a clutch pedal and operation of a gearshift. The first driver model is a model simulating the operation of the clutch pedal of an exemplary driver driving the MT vehicle.

[0012] When the first operation mode is selected, the processor executes the following first to fifth processes. The first process is a process of receiving an operation amount of the accelerator pedal as an input of an operation amount of the gas pedal with respect to the MT vehicle model. The second process is a process of receiving an operation amount of the pseudo-clutch pedal as an input of an operation amount of the clutch pedal with respect to the MT vehicle model. The third process

is a process of receiving a shift position of the pseudo-gearshift as an input of a shift position of the gearshift with respect to the MT vehicle model. The fourth process is a process of calculating the driving wheel torque determined from the operation amount of the gas pedal, the operation amount of the clutch pedal and the shift position of the gearshift using the MT vehicle model. Then, the fifth process is a process of calculating the motor torque for giving the driving wheel torque to driving wheels of the electric vehicle.

[0013] When the second operation mode is selected, the processor executes the following sixth to twelfth processing. The sixth process is a process of disabling the operation of the pseudo-clutch pedal. The seventh process is a process of calculating the operation amount of the clutch pedal using the first driver model. The eighth process is a process of receiving the operation amount of the accelerator pedal as the input of the operation amount of the gas pedal with respect to the MT vehicle model. The ninth process is a process of receiving the operation amount of the clutch pedal calculated using the first driver model as the input of the operation amount of the clutch pedal with respect to the MT vehicle model. The tenth process is a process of receiving the shift position of the pseudo-gearshift as the input of the shift position of the gearshift with respect to the MT vehicle model. The eleventh process is a process of calculating the driving wheel torque determined from the operation amount of the gas pedal, the operation amount of the clutch pedal and the shift position of the gearshift using the MT vehicle model. Then, the twelfth process is a process of calculating the motor torque for giving the driving wheel torque to driving wheels of the electric vehicle.

[0014] According to the electric vehicle according to the present disclosure, when the first operation mode is selected, the operation of the pseudo-clutch pedal and the operation of the pseudo-gearshift are reflected in the calculation of the motor torque using the MT vehicle model, so that the driver can enjoy clutch operation and gearshift operation like an MT vehicle. When the operation mode is switched to the second operation mode, the operation amount of the clutch pedal is calculated by the first driver model, so that the operation of the pseudo clutch pedal is not required, and the driver does not need to feel difficulties peculiar to the MT vehicle. This allows the driver to enjoy driving feeling like the MT vehicle by the electric vehicle without experiencing difficulties peculiar to the MT vehicle.

[0015] In the electric vehicle according to the present disclosure, the operation mode selector may further comprise as the operation mode a third operation mode in addition to the first operation mode and the second operation mode. The third operation mode is an operation mode not requiring the operation of the pseudo-clutch pedal and the operation of the pseudo-gearshift for the control of the electric motor. The memory may also further store a second driver model in addition to the MT vehicle model and the first driver model. The second driver model is a model simulating the operation of the clutch pedal and the operation of the gearshift of the exemplary driver driving the MT vehicle.

[0016] When the third operation mode is selected, the processor may perform the following thirteenth to nineteenth processes. The thirteenth process is a process of disabling the operation of the pseudo-clutch pedal and the operation of the pseudo-gearshift. The fourteenth process is a process of calculating the operation amount of the clutch pedal and the shift position of the gearshift using the second driver model. The fifteenth process is a process of receiving the operation amount of the accelerator pedal as the input of the operation amount of the gas pedal with respect to the MT vehicle model. The sixteenth process is a process of receiving the operation amount of the clutch pedal calculated using the second driver model as the input of the operation amount of the clutch pedal with respect to the MT vehicle model. The seventeenth process is a process of receiving the shift position of the gearshift calculated using the second driver model as the input of the shift position of the gearshift with respect to the MT vehicle model. The eighteenth process is process of calculating the driving wheel torque determined from the operation amount of the gas pedal, the operation amount of the clutch pedal and the shift position of the gearshift using

the MT vehicle model. Then, the nineteenth process is a process of calculating the motor torque for giving the driving wheel torque to driving wheels of the electric vehicle.

[0017] When the operation mode is switched to the third operation mode, the operation amount of the clutch pedal and the shift position of the gearshift are calculated by the second driver model. This makes it needless to operate the pseudo-clutch pedal and the pseudo-gearshift. Therefore, the driver is released from not only clutch operation but also shift operation, which eliminates the difficulties peculiar to the MT vehicle.

[0018] The operation mode selector may be configured to switch the operation mode in accordance with a selection by the driver. Further, the operation mode selector may be configured to switch the operation mode automatically according to a driving scene in accordance with a predetermined rule. Furthermore, the operation mode selector may be configured to learn a relation between the selection by the driver and a driving scene at a time when the operation mode is switched by the driver, and switch the operation mode automatically according to the driving scene in accordance with the relation that is learned.

[0019] As described above, according to the present disclosure, it is possible to provide an electric vehicle capable of enjoying a driving feeling like an MT vehicle without experiencing difficulties peculiar to the MT vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. 1 is a diagram schematically illustrating a configuration of a power system of an electric vehicle according to an embodiment of the present disclosure.

[0021] FIG. 2 is a diagram schematically illustrating a configuration of an information collection system of the electric vehicle according to the embodiment of the present disclosure;

[0022] FIG. 3 is a block diagram illustrating a configuration of a control system of the electric vehicle shown in FIGS. 1 and 2.

[0023] FIG. 4 is a block diagram illustrating functions of the controller of the electric vehicle shown in FIGS. 1 and 2.

[0024] FIG. 5 is a block diagram illustrating an example of an MT vehicle model provided by the controller shown in FIG. 4.

[0025] FIG. 6 is a diagram illustrating an example of an engine model constituting the MT vehicle model shown in FIG. 5.

[0026] FIG. 7 is a diagram illustrating an example of a clutch model constituting the MT vehicle model shown in FIG. 5.

[0027] FIG. 8 is a diagram illustrating an example of an MT model constituting the MT vehicle model shown in FIG. 5.

[0028] FIG. 9 is a diagram illustrating a torque characteristic of an electrical motor realized by motor control using the MT vehicle model in comparison with a torque characteristic of the electric motor realized by normal motor control for an EV.

[0029] FIG. 10 is a diagram illustrating an example of a function of an operation mode selector provided in the controller shown in FIG. 4

[0030] FIG. 11 is a diagram illustrating a function of the driver model provided in the controller shown in FIG. 4 in a clutch operation agent mode.

[0031] FIG. 12 is a diagram illustrating a function of the driver model provided in the controller shown in FIG. 4 in a shift and clutch operation agent mode.

DETAILED DESCRIPTION

[0032] Hereunder, embodiments of the present disclosure will be described with reference to the drawings. Note that when the numerals of numbers, quantities, amounts, ranges and the like of

respective elements are mentioned in the embodiments shown as follows, the present disclosure is not limited to the mentioned numerals unless specially explicitly described otherwise, or unless the disclosure is explicitly designated by the numerals theoretically. Furthermore, structures and steps that are described in the embodiments shown as follows are not always indispensable to the disclosure unless specially explicitly shown otherwise, or unless the disclosure is explicitly designated by the structures or the steps theoretically.

1. Configuration of Electric Vehicle

[0033] FIG. 1 is a diagram schematically illustrating a configuration of a power system of an electric vehicle **10** according to the present embodiment. As shown in FIG. 1, the electric vehicle **10** is provided with an electric motor **2** as a power source. The electric motor **2** is, for example, a brushless DC motor or a three-phase AC synchronous motor. The electric motor **2** is provided with a rotation speed sensor **40** for detecting its rotation speed. An output shaft **3** of the electric motor **2** is connected to one end of a propeller shaft **5** via a gear mechanism **4**. The other end of the propeller shaft **5** is connected to a drive shaft **7** at the front of the vehicle via a differential gear **6**. [0034] The electric vehicle **10** includes driving wheels **8**, which are front wheels, and driven wheels **12**, which are rear wheels. The driving wheels **8** are provided on both ends of the drive shaft **7**, respectively. Each wheel **8** and **12** is provided with a wheel speed sensor **30**. In FIG. 1, only the wheel speed sensor **30** of the right-hand rear wheel is represented. The wheel speed sensor **30** is also used as a vehicle speed sensor for detecting the vehicle speed of the electric vehicle **10**. The wheel speed sensor **30** is connected to a controller **50** to be described later by an in-vehicle network such as CAN (Controller Area Network).

[0035] The electric vehicle **10** includes a battery **14** and an inverter **16**. The battery **14** stores electrical energy that drives the electric motor **2**. The inverter **16** converts DC power input from the battery **14** to driving power for the electric motor **2**. Power conversion by the inverter **16** is performed by PWM (Pulse Wave Modulation) control by the controller **50**. The inverter **16** is connected to the controller **50** by the in-vehicle network.

[0036] The electric vehicle **10** includes an accelerator pedal **22** for inputting an acceleration request and a brake pedal **24** for inputting a braking request as operation request input devices for inputting operation requests from the driver to the electric vehicle **10**. The accelerator pedal **22** is provided with an accelerator position sensor **32** for detecting an accelerator opening P_{ap} [%] which is an operation amount of the accelerator pedal **22**. The brake pedal **24** is provided with a brake position sensor **34** for detecting a brake depression amount which is an operation amount of the brake pedal **24**. The accelerator position sensor **32** and the brake position sensor **34** are connected to the controller **50** by the in-vehicle network.

[0037] The electric vehicle **10** further includes a pseudo-gearshift **26** and a pseudo-clutch pedal **28** as operation inputting devices. A gearshift and a clutch pedal are devices that operate a manual transmission (MT), but of course the electric vehicle **10** is not equipped with the MT. The pseudo-gearshift **26** and the pseudo-clutch pedal **28** are dummies that differ from the original gearshift and clutch pedal.

[0038] The pseudo-gearshift **26** has a structure that simulates a gearshift installed in an MT vehicle. The arrangement and operating feeling of the pseudo-gearshift **26** are equivalent to those of the real MT vehicle. The pseudo-gearshift **26** has positions that correspond to each gear stage, for example, first-speed stage, second-speed stage, third-speed stage, fourth-speed stage, fifth-speed stage, sixth-speed stage, reverse stage, and neutral stage. The pseudo-gearshift **26** is equipped with a shift position sensor **36** for detecting gear stage by determining which position the pseudo-gearshift **26** is in. The shift position sensor **36** is connected to the controller **50** by the in-vehicle network.

[0039] The pseudo-clutch pedal **28** has a structure that simulates a clutch pedal installed in the MT vehicle. The arrangement and operating feeling of the pseudo-clutch pedal **28** are equivalent to those of the real MT vehicle. When the driver wants to change the setting of the gear stage by the pseudo-gearshift **26**, the driver depresses the pseudo-clutch pedal **28**, and after finishing the setting

change of the gear stage, ceases depressing to release the pseudo-clutch pedal **28**. The pseudo-clutch pedal **28** is equipped with a clutch position sensor **38** for detecting depression amount P_c [%] of the pseudo-clutch pedal **28**. The clutch position sensor **38** is connected to the controller **50** by the in-vehicle network.

[0040] The electric vehicle **10** is equipped with a pseudo-engine speed meter **44**. An engine speed meter is a device that displays a rotation speed of an internal combustion engine to the driver, but of course, the electric vehicle **10** is not equipped with the internal combustion engine. The pseudo-engine speed meter **44** is, of course, a dummy that differs from the original engine speed meter. The pseudo-engine speed meter **44** has a structure that simulates the engine speed meter installed in the conventional vehicle. The pseudo-engine speed meter **44** may be a mechanical type, a liquid crystal display type, or a projection display type using a head-up display. In the case of the liquid crystal display type and the projection display type, a revolution limit may be arbitrarily set in the pseudo-engine speed meter **44**. The pseudo-engine speed meter **44** is connected to the controller **50** by the in-vehicle network.

[0041] The electric vehicle **10** is equipped with an HMI (Human Machine Interface) unit **42**. The driver may use the HMI unit **42** to select an operation mode relating to the operation of the pseudo-gearshift **26** and the pseudo-clutch pedal **28**. In this case, the HMI unit **42** functions as an operation mode selector. Details will be described later, there are four operation modes selectable on the HMI unit **42**: a manual operation mode, a clutch operation agent mode, a shift and clutch operation agent mode, and an automatic selection mode. The HMI unit **42** may be, for example, a touch panel type HMI or a voice interactive HMI. The HMI unit **42** is connected to the controller **50** by the in-vehicle network.

[0042] FIG. **2** is a diagram schematically illustrating a configuration of an information collection system of the electric vehicle **10** according to the present embodiment. As shown in FIG. **2**, the electric vehicle **10** includes a camera **60**, a radar **62**, and a LIDAR **64** as external sensors for detecting conditions in which the self vehicle is placed. The camera **60** is mounted so as to image at least the front of the self vehicle. The camera **60** is, for example, a stereo camera equipped with a CMOS image sensor. The radar **62** is specifically a millimeter wave radar and is attached to the front portion of the vehicle. The LIDAR **64** is mounted, for example, on the roof of the vehicle. Further, the electric vehicle **10** is provided with a navigation device **66** for estimating the current location of the self vehicle on a map based on map information and GPS information. The external sensors **60**, **62**, **64** and the navigation device **66** are connected to the controller **50** by the in-vehicle network.

[0043] The controller **50** is typically an ECU (Electronic Control Unit) mounted on the electric vehicle **10**. The controller **50** may be a combination of a plurality of ECUs. The controller **50** includes an interface **52**, a memory **54**, and a processor **56**. The in-vehicle network is connected to the interface **52**. The memory **54** includes a RAM (Random Access Memory) for temporarily recording data and a ROM (Read Only Memory) for storing a control program executable by the processor **56** and various data related to the control program. The processor **56** executes the control program read with the related data from the memory **54**, and generates a control signal based on the signal obtained from each sensor.

[0044] FIG. **3** is a block diagram illustrating a configuration of a control system of the electric vehicle **10** according to the present embodiment. The controller **50** receives signals at least from the wheel speed sensor **30**, the accelerator position sensor **32**, the brake position sensor **34**, the shift position sensor **36**, the clutch position sensor **38**, the rotation speed sensor **40**, the HMI unit **42**, the camera **60**, the radar **62**, the LIDAR **64**, and the navigation device **66**. The in-vehicle network is used for communication between these sensors and the controller **50**. Although not shown, in addition to these sensors, various other sensors are mounted on the electric vehicle **10**, and connected to the controller **50** by the in-vehicle network.

[0045] Further, from the controller **50**, a signal is output to at least the inverter **16** and the pseudo-

engine speed meter **44**. The in-vehicle network is used for communication between these devices and the controller **50**. Although not shown, in addition to these devices, various other actuators and indicators are mounted on the electric vehicle **10**, and connected to the controller **50** by the in-vehicle network.

[0046] The controller **50** has a function as an operation mode switching unit **500** and a function as a control signal calculation unit **520**. More specifically, the processor **56** functions at least as the operation mode switching unit **500** and the control signal calculation unit **520** when a program stored in the memory **54** is executed by the processor **56**. The operation mode selection is a function to determine which operation mode is selected between a manual operation mode, the clutch operation agent mode, and the shift and clutch operation agent mode. The control signal calculation is a function to calculate a control signal for an actuator or a device. The control signal includes at least a signal for PWM control of the inverter **16**, and a signal for displaying information on the pseudo-engine speed meter **44**. These functions of the controller **50** will be described below.

2. Functions of Controller

2-1. Motor Torque Calculation Function

[0047] FIG. **4** is a block diagram illustrating functions of the controller **50** according to the present embodiment, in particular, a function relating to a calculation of the motor torque command value for the electric motor **2**. The controller **50** calculates the motor torque command value by the function shown in this block diagram, and generates the control signal for the PWM control of the inverter **16** based on the motor torque command value.

[0048] As shown in FIG. **4**, the control signal calculation unit **520** comprises an MT vehicle model **530**, a required motor torque calculation unit **540**, and a driver model **550**. The control signal calculation unit **520** receives signals from the wheel speed sensor **30**, the accelerator position sensor **32**, the shift position sensor **36**, the clutch position sensor **38**, and the rotation speed sensor **40**. The control signal calculation unit **520** processes the signals from these sensors and calculates motor torque which the electric motor **2** is made to output.

[0049] The driving wheel torque of the MT vehicle is determined from the operation of a gas pedal that controls fuel supply to the engine, the operation of a gearshift that switches a gear stage of the MT, and the operation of a clutch pedal that operates a clutch between the engine and the MT. The MT vehicle model **530** is a model that calculates the driving wheel torque obtained by operating the accelerator pedal **22**, the pseudo-clutch pedal **28**, and the pseudo-gearshift **26** assuming that the electric vehicle **10** is equipped with the engine, the clutch, and the MT. Hereinafter, the engine, the clutch, and the MT, which are imaginarily realized by the MT vehicle model **530** in the MT mode, will be referred to as an imaginary engine, an imaginary clutch, and an imaginary MT.

[0050] The MT vehicle model **530** receives an accelerator opening P_{ap} detected by the accelerator position sensor **32** as an operation amount of the gas pedal of the imaginary engine. A shift position Spd detected by the shift position sensor **36** is input to the MT vehicle model **530** as a shift position of the gearshift of the imaginary MT. Further, A clutch pedal depression amount P_{cd} detected by the clutch position sensor **38** is input to the MT vehicle model **530** as an operation amount of the clutch pedal of the imaginary clutch. The MT vehicle model **530** also receives a vehicle speed V_w (or wheel speed) detected by the wheel speed sensor **30** as a signal indicating the load condition of the vehicle. The detail of the MT vehicle model **530** will be described later.

[0051] The required motor torque calculation unit **540** converts the driving wheel torque T_w calculated by the MT vehicle model **530** into a required motor torque T_m . The required motor torque T_m is the motor torque required for realizing the driving wheel torque T_w calculated by the MT vehicle model **530**. The reduction ratio from the output shaft **3** of the electric motor **2** to the driving wheels **8** is used to convert the driving wheel torque T_w into the required motor torque T_m .

[0052] The driver model **550** is a model simulating the operation of the clutch pedal or the operation of the gearshift and the clutch pedal of an exemplary driver driving the MT vehicle. The

exemplary driver is an imaginary driver that serves as a model for operating the MT vehicle. The driver model **550** receives the shift position Spd detected by the shift position sensor **36** and the clutch pedal depression amount Pcd detected by the clutch position sensor **38**. The vehicle speed Vw (or wheel speed) detected by the wheel speed sensor **30**, the accelerator opening Pap detected by the accelerator position sensor **32**, and the imaginary engine speed Ne calculated by the MT vehicle model **530** are also input to the driver model **550**. The shift position Spd input to the MT vehicle model **530** can be switched to the shift position Spm calculated by the driver model **550**. Further, the clutch pedal depression amount Pcd input to the MT vehicle model **530** can be switched to the clutch pedal depression amount Pcm calculated by the driver model **550**. Details of the driver model **550** will be described later.

[0053] The driver model **550** receives a mode switching signal from the operation mode switching unit **500**. The mode switching signal is a signal for switching a shift position signal input from the driver model **550** to the MT vehicle model **530** between the shift position Spd and the shift position Spm. The mode switching signal is also a signal for switching a clutch position signal input from the driver model **550** to the MT vehicle model **530** between the clutch pedal depression amount Pcd and the clutch pedal depression amount Pcm. The operation mode switching unit **500** receives signals from the HMI unit **42**, the navigation device **66**, and the external sensors **60**, **63**, **64**. The operation mode switching unit **500** determines the mode switching signal to be output to the driver model **550** based on these signals. Details of the operation mode switching unit **500** will be described later.

2-2. MT Vehicle Model

2-2-1. Summary

[0054] Next, the MT vehicle model **530** will be described. FIG. 5 is a block diagram illustrating an example of the MT vehicle model **530**. The MT vehicle model **530** comprises an engine model **531**, a clutch model **532**, an MT model **533**, and an axles and drive wheels model **534**. The engine model **531** is a model of the imaginary engine. The clutch model **532** is a model of the imaginary clutch. The MT model **533** is a model of the imaginary MT. The axles and drive wheels model **534** is a model of the imaginary torque transmission system from the axles to the driving wheels. Each model may be represented by a calculation formula or may be represented by a map.

[0055] Calculation results are input and output between models. Further, the accelerator opening Pap detected by the accelerator position sensor **32** is input to the engine model **531**. A clutch pedal depression amount Pc is input to the clutch model **532**. The clutch pedal depression amount Pc is the clutch pedal depression amount Pcd detected by the clutch position sensor **38** or the clutch pedal depression amount Pcm calculated by the driver model **550**. A shift position Sp is input to the MT model **533**. The shift position Sp is the shift position Spd detected by the shift position sensor **36** or the shift position Spm calculated by the driver model **550**. Furthermore, in the MT vehicle model **530**, the vehicle speed Vw (or wheel speed) detected by the wheel speed sensor **30** is used in a plurality of models. In the MT vehicle model **530**, a driving wheel torque Tw and an imaginary engine speed Ne are calculated based on these input signals.

2-2-2. Engine Model

[0056] The engine model **531** calculates the imaginary engine speed Ne and an imaginary engine output torque Teout. The engine model **531** comprises a model to calculate the imaginary engine speed Ne and a model to calculate the imaginary engine output torque Teout. For calculating the imaginary engine speed Ne, for example, a model expressed by the following equation (1) is used. In the following equation (1), the imaginary engine speed Ne is calculated from a rotation speed Nw of the wheel **8**, a total reduction ratio R, and a slip ratio Rslip of the imaginary clutch mechanism.

$$[00001] \quad N_e = N_w \times \frac{1}{R} \times R_{\text{slip}} \quad (1)$$

[0057] In the equation (1), the rotation speed Nw of the wheel **8** is detected by the wheel speed

sensor **30**. The total reduction ratio R is calculated from a gear ratio r calculated by the MT model **533** to be described later and the reduction ratio specified by the axles and drive wheels model **534**. The slip ratio R_{slip} is calculated by the clutch model **532** to be described later. The imaginary engine speed N_e is displayed on the pseudo-engine speed meter **44** when the MT mode is selected. [0058] However, the equation (1) is an equation for calculating the imaginary engine speed N_e in a condition where the imaginary engine and the imaginary MT are connected by the imaginary clutch mechanism. When the imaginary clutch mechanism is disengaged, the imaginary engine torque T_e generated in the imaginary engine can be regarded as being used to increase the imaginary engine speed N_e . The imaginary engine torque T_e is a torque obtained by adding the torque due to the moment of inertia to the imaginary engine output torque T_{eout} . When the imaginary clutch mechanism is disengaged, the imaginary engine output torque T_{eout} is zero. Therefore, when the imaginary clutch mechanism is disengaged, the engine model **531** calculates the imaginary engine speed N_e by the following equation (2) using the imaginary engine torque T_e and the moment of inertia J of the imaginary engine. For the calculation of the imaginary engine torque T_e , a map with the accelerator opening P_{ap} as a parameter is used.

$$[00002] J \times \frac{30}{\pi} \times \frac{d}{dt} N_e = T_e \quad (2)$$

[0059] Incidentally, during idling of the MT vehicle, idle speed control (ISC control) is executed to maintain the engine speed at a constant rotation speed. Therefore, the engine model **531** calculates the imaginary engine speed N_e as a predetermined idling speed (for example, 1000 rpm), when the imaginary clutch mechanism is disengaged, the vehicle speed is 0, and the accelerator opening P_{ap} is 0%. When the driver depresses the accelerator pedal **22** to perform racing while the vehicle is stopped, the idling speed is used as the initial value of the imaginary engine speed N_e calculated by equation (2).

[0060] The engine model **531** calculates the imaginary engine output torque T_{eout} from the imaginary engine speed N_e and the accelerator opening P_{ap} . For calculating the imaginary engine output torque T_{eout} , for example, a two-dimensional map as shown in FIG. **6** is used. This two-dimensional map is a map defining the relationship between the accelerator opening P_{ap} in steady-state, the imaginary engine speed N_e , and the imaginary engine output torque T_{eout} . In this map, imaginary engine outputting torque T_{eout} for imaginary engine speed N_e is given for each accelerator opening P_{ap} . The torque characteristic shown in FIG. **7** can be set to the characteristic assumed for a gasoline engine or can be set to that assumed for a diesel engine. In addition, the torque characteristic can be set to that assumed for a natural intake engine or can be set to that assumed for a turbocharged engine. A selector switching the imaginary engine of the MT mode may be installed so that the driver can chose a preferred setting. The imaginary engine output torque T_{eout} calculated by the engine model **531** is output to the clutch model **532**.

2-2-3. Clutch Model

[0061] The clutch model **532** calculates a torque transmission gain k . The torque transmission gain k is a gain for calculating the torque transmission degree of the imaginary clutch corresponding to the depression amount of the pseudo-clutch pedal **28**. The clutch model **532** has, for example, a map as shown in FIG. **7**. In this map, the torque transmission gain k is given for the clutch pedal depression amount P_c . In FIG. **7**, the torque transmission gain k is 1 when the clutch pedal depression amount P_c is in the range from P_{c0} to P_{c1} , the torque transmission gain k monotonically decreases at a constant slope when the clutch pedal depression amount P_c is in the range from P_{c1} to P_{c2} , and the torque transmission gain k is 0 when the clutch pedal depression amount P_c is in the range from P_{c2} to P_{c3} . Here, P_{c0} corresponds to the position where the clutch pedal depression amount P_c is 0%, P_{c1} corresponds to the position of the play limit when the clutch pedal is depressed, P_{c3} corresponds to the position where the clutch pedal depression amount P_c is 100%, and P_{c2} corresponds to the play limit when the clutch pedal is returned from P_{c3} .

[0062] The map shown in FIG. 7 is an example. The change in the torque transmission gain k with respect to an increase in the clutch pedal depression amount P_c is not limited to the change curve shown in FIG. 8 as long as it is a broad monotonic decrease toward 0. For example, the change in the torque transmission gain k in the range from P_{c1} to P_{c2} may be a monotonically decreasing curve that is convex upward or a monotonically decreasing curve that is convex downward.

[0063] The clutch model 532 calculates a clutch output torque T_{cout} using the torque transmission gain k . The clutch output torque T_{cout} is the torque output from the imaginary clutch. The clutch model 532 calculates the clutch output torque T_{cout} from the imaginary engine output torque T_{eout} and the torque transmission gain k by, for example, the following equation (3). The clutch output torque T_{cout} calculated by the clutch model 532 is output to the MT model 533.

$$[00003] \quad T_{cout} = T_{eout} \times k \quad (3)$$

[0064] Further, the clutch model 532 calculates the slip ratio R_{slip} . The slip ratio R_{slip} is used to calculate the imaginary engine speed N_e in the engine model 531. The slip ratio R_{slip} can be calculated by using a map in which the slip ratio R_{slip} is given to the clutch pedal depression amount P_c , in the same manner as the torque transmission gain k . Instead of such a map, the slip ratio R_{slip} may be calculated from the torque transmission gain k by the following equation (4) representing a relation between the slip ratio R_{slip} and the torque transmission gain k .

$$[00004] \quad R_{slip} = 1 - k \quad (4)$$

2-2-4. MT Model

[0065] The MT model 533 calculates the gear ratio r . The gear ratio r is the gear ratio determined from the shift position S_p of the pseudo-gearshift 26 in the imaginary MT. The shift position S_p of the pseudo-gearshift 26 and the gear stage of the imaginary MT are in a one-to-one relation. The MT model 533 has, for example, a map as shown in FIG. 8. In this map, the gear ratio r is given for the gear stage. As shown in FIG. 8, the larger gear stage, the smaller the gear ratio r .

[0066] The MT model 533 calculates a transmission output torque T_{gout} using the gear ratio r . The transmission output torque T_{gout} is the torque output from the imaginary transmission. The MT model 533 calculates the transmission output torque T_{gout} from the clutch output torque T_{cout} and the gear ratio r by, for example, the following equation (5). The transmission output torque T_{gout} calculated by the MT model 533 is output to the axles and drive wheels model 534.

$$[00005] \quad T_{gout} = T_{cout} \times r \quad (5)$$

2-2-5. Axles and Drive Wheels Model

[0067] The axles and drive wheels model 534 calculates the driving wheel torque T_w using a predetermined reduction ratio rr . The reduction ratio rr is a fixed value determined by the mechanical structure from the imaginary MT to the driving wheels 8. The value obtained by multiplying the reduction ratio rr by the gear ratio r is the total reduction ratio R described above. The axles and drive wheels model 534 calculates the driving wheel torque T_w from the transmission output torque T_{gout} and the reduction ratio rr by, for example, the following equation (6). The driving wheel torque T_w calculated by the axles and drive wheels model 534 is output to the required motor torque calculation unit 540.

$$[00006] \quad T_w = T_{gout} \times rr \quad (6)$$

2-2-6. Torque Characteristic of Electric Motor Realized by Motor Control Using MT Vehicle Model

[0068] The required motor torque calculation unit 540 converts the driving wheel torque T_w calculated by the MT vehicle model 530 into motor torque. FIG. 9 is a diagram illustrating the torque characteristic of the electrical motor 2 realized by motor control using the MT vehicle model 530 in comparison with the torque characteristic of the electric motor realized by normal motor control for the EV. According to the motor control using the MT vehicle model 530, as shown in FIG. 9, it is possible to realize a torque characteristic (solid line in the drawing) such as to simulate the torque characteristic of the MT vehicle according to the gear stage set by the pseudo-gearshift

26.

2-3. Operation Mode Selection Function

2-3-1. Summary of Each Operation Mode

[0069] According to the MT vehicle model **530** described above, by reflecting the operation of the pseudo-gearshift **26** and the pseudo-clutch pedal **28** by the driver to the motor torque, the driver can drive the electric vehicle **10** like the MT vehicle. However, for the driver who is not familiar with the operation of the MT vehicle, there are many scenes in which the driver feels difficulty in the clutch operation and the shift operation. Further, even if the driver is familiar with the operation of the MT vehicle, there is a possibility that the driver may want to drive easily without the clutch operation or the shift operation. As a countermeasure to such a problem, the controller **50** is provided with a function of performing the clutch operation and the shift operation on behalf of the driver. Specifically, the following three operation modes are provided as modes of the clutch operation and the shift operation by the driver.

[0070] First, the first operation mode provided by the controller **50** is a mode in which the operation of the pseudo-clutch pedal **28** and the operation of the pseudo-gearshift **26** are reflected in the control of the electric motor **2**. In the first operation mode, the driver can control the motor torque by the operation of the pseudo-clutch pedal **28** and the operation of the pseudo-gearshift **26** to drive the electric vehicle **10** like the MT vehicle. Hereinafter, the first operation mode is referred to as the manual operation mode.

[0071] The second operation mode provided by the controller **50** is a mode in which the operation of the pseudo-clutch pedal **28** is not required in the control of the electric motor **2**. In the second operation mode, the clutch operation required to drive the MT vehicle is performed by the controller **50** on behalf of the driver. Hereinafter, the second operation mode is referred to as the clutch operation agent mode.

[0072] The third operation mode is a mode in which the operation of the pseudo-gearshift **26** and the operation of the pseudo-clutch pedal **28** are not required in the control of the electric motor **2**. In the third operation mode, both the clutch operation and the shift operation required to drive the MT vehicle are performed by the controller **50** on behalf of the driver. Hereinafter, the third operation mode is referred to as the shift and clutch operation agent mode.

2-3-2. Operation Mode Switching Unit

[0073] Switching of the above three operation modes is performed by the operation mode switching unit **500**. The operation mode switching unit **500** includes a mode switching signal output unit **502** and an automatic switching determination unit **504**. The mode switching signal output unit **502** outputs the mode switching signal corresponding to the operation mode selected by the HMI unit **42** to the driver model **550**. The automatic switching determination unit **504** functions when the operation mode selected by the HMI unit **42** is the automatic switching mode. The signal from the navigation device **66** and the signals from the external sensors **60**, **63**, **64** are input to the automatic switching determination unit **504**.

[0074] The mode switching signal output unit **502** sets a manual operation flag **502a** when the manual operation mode is selected by the HMI unit **42**. When the manual operation flag **502a** is set, a mode switching signal instructing switching to the manual operation mode is input from the mode switching signal output unit **502** to the driver model **550**.

[0075] The mode switching signal output unit **502** sets a clutch operation agent flag **502b** when the clutch operation agent mode is selected by the HMI unit **42**. When the clutch operation agent flag **502b** is set, a mode switching signal instructing switching to the clutch operation agent mode is input from the mode switching signal output unit **502** to the driver model **550**.

[0076] The mode switching signal output unit **502** sets a shift and clutch operation agent flag **502c** when the shift and clutch operation agent mode is selected with the HMI unit **42**. When the shift and clutch operation agent flag **502c** is set, a mode switching signal instructing switching to the shift and clutch operation agent mode is input from the mode switching signal output unit **502** to

the driver model **550**.

2-3-3. Automatic Switching of Operation Mode According to Driving Scene

[0077] The automatic switching determination unit **504** includes a driving scene registration unit **504a**. In the driving scene registration unit **504a**, a typical scene in which the driver wants the clutch operation to be performed on behalf of the driver and a typical scene in which the driver wants both the shift operation and the clutch operation to be performed on behalf of the drive are stored in advance. When the automatic switching mode is selected by the HMI unit **42**, the automatic switching determination unit **504** automatically selects the operation mode suitable for the current driving scene among the manual operation mode, the clutch operation agent mode, and the shift and clutch operation agent mode.

[0078] Examples of the scenes in which the driver wants the clutch operation to be performed on behalf of the driver are as follows: reverse, a sharp curve, a traffic jam, a railroad crossing, a garage, and a gravel road. In the driving scene registration unit **504a**, these scenes are registered in association with the clutch operation agent mode. And, a traffic jam is an example of the scene in which the driver wants both the shift operation and the clutch operation to be performed on behalf of the drive. In the driving scene registration unit **504a**, such a scene is registered in association with the shift and clutch operation agent mode. The automatic switching determination unit **504** selects an operation mode suitable for the current driving scene in accordance with the data registered in the driving scene registration unit **504a**, and inputs the selected operation mode to the mode switching signal output unit **502**. Incidentally, the manual operation mode is selected as the basic mode in the automatic switching determination unit **504**. Switching of the operation mode from the manual operation mode to the clutch operation agent mode or from the manual operation mode to the shift and clutch operation agent mode is performed only in the scene registered in the driving scene registration unit **504a**.

2-3-4. Machine Learning of Operation Mode

[0079] The automatic switching determination unit **504** includes a machine learning unit **504b**. When the driver selects the manual operation mode, the clutch operation agent mode, or the shift and clutch operation agent mode with the HMI unit **42**, the machine learning unit **504b** identifies the driving scene in which the selection has been made. For the identification of the driving scene, the location information of the self vehicle on the map obtained by the navigation device **66**, and the peripheral information of the self vehicle obtained by the external sensors **60**, **62**, and **64** are used. Based on these information, the machine learning unit **504b** identifies what kind of driving scene the vehicle is in at present.

[0080] The machine learning unit **504b** performs machine learning to learn the relationship between the operation mode selected by the driver with the HMI unit **42** and the driving scene at the selected time. For example, supervised learning is used for the machine learning by the machine learning unit **504b**. In the supervised learning, the teacher data that uses the operation mode selected with the HMI unit **42** as an output and the driving scene at the selected time as an input is used. As the algorithm of the supervised learning, k-neighborhood method, decision tree, random forest, support vector machine, logistic regression, and neural network including deep learning, etc. can be used.

2-4. Driver Model

2-4-1. Functions in Manual Operation Mode

[0081] Next, the driver model **550** will be described. The function of the driver model **550** depends on the operation mode. In the manual operation mode, which is the basic operation mode, the driver model **550** does not function. In FIG. **4**, the shift position Spd input from the shift position sensor **36** to the driver model **550** is input from the driver model **550** to the MT vehicle model **530** as it is. Further, the clutch pedal depression amount Pcd input from the clutch position sensor **38** to the driver model **550** is input from the driver model **550** to the MT vehicle model **530** as it is. That is, in the manual operation mode, the operation of the pseudo-clutch pedal **28** and the operation of

the pseudo-gearshift **26** are reflected in the calculation of the motor torque using the MT vehicle model **530**. This allows the driver to enjoy the clutch operation and the shift operation like the MT vehicle.

2-4-2. Functions in Clutch Operation Agent Mode

[0082] When switching from the manual operation mode to the clutch operation agent mode is instructed by the operation mode switching unit **500**, the driver model **550** functions as a first driver model **550a** shown in FIG. **11**. In the first driver model **550a**, the signal from the clutch position sensor **38** is not used, and the operation of the pseudo-clutch pedal **28** by the driver is disabled.

[0083] The first driver model **550a** includes a clutch operation model **552**. The clutch operation model **552** is a model simulating the clutch operation of the exemplary driver. The clutch operation model **552** receives the vehicle speed V_w (or wheel speed) detected by the wheel speed sensor **30**, the imaginary engine speed N_e , and a signal from the shift position sensor **36**.

[0084] In the clutch operation model **552**, the signal from the shift position sensor **36** is used to time the clutch operation. The clutch operation model **552** sets the clutch pedal depression amount P_{cm} input to the MT vehicle model **530** to a maximum value so as to disengage the imaginary clutch mechanism when the shift operation of the driver is detected from the signal from the shift position sensor **36**.

[0085] In the clutch operation model **552**, the vehicle speed V_w and the imaginary engine speed N_e are used to calculate the clutch pedal depression amount P_{cm} . In order to smoothly match a rotation speed of an input shaft of the imaginary MT calculated from the vehicle speed V_w and the imaginary engine speed N_e , the clutch operation model **552** calculates the clutch pedal depression amount P_{cm} based on a rotation speed difference between the rotation speed of the input shaft of the imaginary MT and the imaginary engine speed N_e .

[0086] As described above, in the clutch operation agent mode, since the clutch pedal depression amount P_{cm} is calculated by the first driver model **550a**, the pseudo-clutch pedal **28** does not need to be operated. Therefore, the driver does not have to experience difficulties peculiar to the MT vehicle with the clutch operation. This allows the driver to enjoy the driving feeling like the MT vehicle in the electric vehicle **10** without any difficulty peculiar to the MT vehicle.

2-4-3. Function in Shift and Clutch Operation Agent Mode

[0087] When switching from the manual operation mode to the shift and clutch operation agent mode is instructed by the operation mode switching unit **500**, the driver model **550** functions as a second driver model **550b** shown in FIG. **12**. In the second driver model **550b**, the signal from the clutch position sensor **38** is not used and the operation of the pseudo-clutch pedal **28** by the driver is disabled. At the same time, no signal from the shift position sensor **36** is used, and the operation of the pseudo-gearshift **26** by the driver is also disabled.

[0088] The second driver model **550b** includes a clutch operation model **552** and a shift operation model **554**. The shift operation model **554** is a model simulating a shift operation of the exemplary driver. The shift operation model **554** receives the vehicle speed V_w (or wheel speed) detected by the wheel speed sensor **30**, and the imaginary engine speed N_e . The clutch operation model **552** is common to that of the first driver model **550a**. However, in the second driver model **550b**, a signal for timing the clutch operation is input to the clutch operation model **552** from the shift operation model **554**.

[0089] The shift operation model **554** uses a shift pattern map that defines the relationship between the vehicle speed V_w , the imaginary engine speed N_e , and the gear stage of the imaginary MT. The shift position S_{pm} input from the shift operation model **554** to the MT vehicle model **530** corresponds one-to-one to the gear stage of the imaginary MT. The shift pattern map is prepared separately for upshifting and for downshifting.

[0090] The shift operation model **554** determines the shift position S_{pm} at accelerating using the shift pattern map for upshifting. The shift pattern map, for example, is configured such that an

upshift is performed when the imaginary engine speed N_e is increased to a predetermined upshift reference speed. The upshift reference speed is the engine speed in front of the torque band in the engine model **531**, and is set for each gear stage.

[0091] The shift operation model **554** determines the shift position S_{pm} at decelerating using the shift pattern map for downshifting. This shift pattern map, for example, is configured such that a downshift is performed when the imaginary engine speed N_e is reduced to a predetermined downshift reference speed. The downshift reference speed is set to a lower engine speed than the upshift reference speed.

[0092] As described above, in the shift and clutch operation agent mode, the shift position S_{pm} and the clutch pedal depression amount P_{cm} are calculated by the second driver model **550b**.

Therefore, the driver does not need to operate the pseudo-clutch pedal **28** and does not need to operate the pseudo-gearshift **26**. Therefore, the driver does not have to experience difficulties peculiar to the MT vehicle with the clutch operation and the shift operation. This allows the driver to enjoy the driving feeling like the MT vehicle in the electric vehicle **10** without any difficulty peculiar to the MT vehicle.

3. Other

[0093] The electric vehicle **10** according to the above embodiment is an FF vehicle that drives the front wheels in one electric motor **2**. However, the present disclosure is also applicable to an electric vehicle in which two electric motor are arranged in front and rear to drive each of the front and rear wheels. The present disclosure is also applicable to an electric vehicle comprising an in-wheel motor on each wheel. The MT vehicle model of these cases may be a model in which an all-wheel-drive vehicle with MT is modeled.

[0094] The electric vehicle **10** according to the above embodiment is not provided with a transmission. However, the present disclosure is also applicable to an electric vehicle having a stepped or continuously variable automatic transmission. In this case, the power train consisting of the electric motor and the automatic transmission may be controlled so as to output the motor torque calculated by MT vehicle model.

[0095] In the above embodiment, the operation mode switching unit **500** constitutes an operation mode selector together with the HMI unit **42**. Although the operation mode switching unit **500** is one function of the controller **50**, all of the functions of the operation mode switching unit **500** may be transferred to the HMI unit **42**. That is, the HMI unit **42** itself may be configured as an operation mode selector.

Claims

1. An electric vehicle configured to use an electric motor as a power device for traveling, the electric vehicle comprising: a controller having an operation mode in which the electric motor is operated to simulate a torque characteristic of a manual transmission vehicle by using a manual transmission model, the manual transmission model being a model of an imaginary manual transmission, wherein the operation mode includes an automatic shift mode in which a gear stage of the imaginary manual transmission is automatically shifted and a manual shift mode in which the gear stage of the imaginary manual transmission is manually shifted by a driver, and the controller automatically switches the operation mode in response to detection of a predetermined condition.
2. The electric vehicle according to claim 1, further comprising: a pseudo-gearshift configured to allow manual shifting of the gear stage of the imaginary manual transmission, wherein the manual shift mode is performed by operating the pseudo-gearshift.
3. The electric vehicle according to claim 2, further comprising: a pseudo-clutch pedal configured to allow manual shifting of the gear stage of the imaginary manual transmission when operated

together with the pseudo-gearshift, wherein the manual shift mode is performed by operating the pseudo-gearshift and the pseudo-clutch pedal.
